Problem Set 3

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Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

(a) Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

Answer: First, we import the data and check the three variables for missing data.

```
incumbents <- read.csv("incumbents_subset.csv", row.names = 1)
incumbents <- incumbents[c("voteshare", "difflog", "presvote")]
print(dim(incumbents))
print(colSums(is.na(incumbents)))
summary(incumbents)
```

```
[1] 3193    3
voteshare   difflog  presvote
        0         0         0
   voteshare         difflog           presvote
 Min.   :0.3476   Min.   :-3.0601   Min.   :0.1931
 1st Qu.:0.5846   1st Qu.: 0.6734   1st Qu.:0.4695
 Median :0.6569   Median : 1.6106   Median :0.5478
 Mean   :0.6552   Mean   : 1.8278   Mean   :0.5512
 3rd Qu.:0.7233   3rd Qu.: 2.9857   3rd Qu.:0.6240
 Max.   :0.9930   Max.   : 5.8558   Max.   :0.9606
```

All 3,193 elections have complete data (no NAs).

We regress incumbent `voteshare` $\sim$ `difflog` (log incumbent spending minus log challenger spending).

```
vote_spending <- lm(voteshare ~ difflog, data = incumbents)
summary(vote_spending)
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.579031   0.002251  257.19   <2e-16 ***
difflog     0.041666   0.000968   43.04   <2e-16 ***

Multiple R-squared:  0.3673,    Adjusted R-squared:  0.3671
```

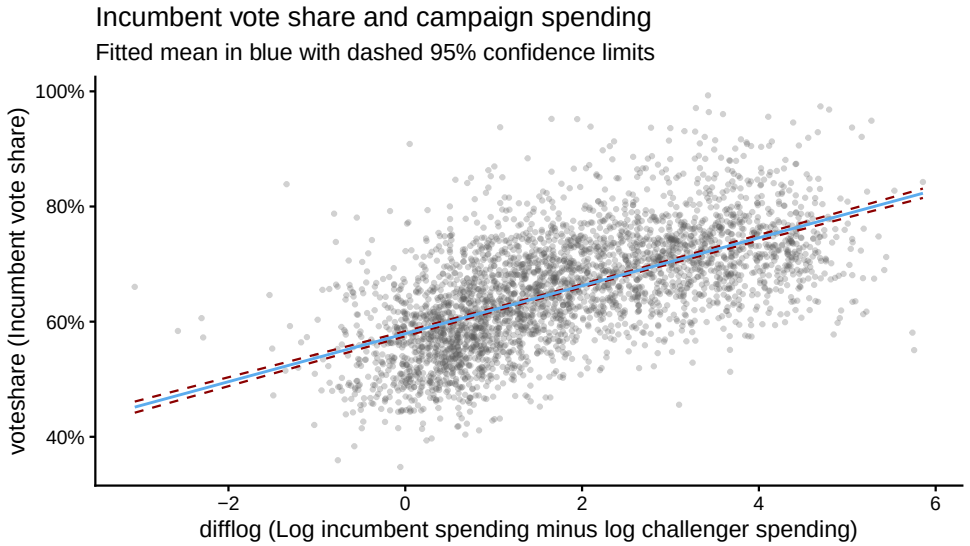
The spending gap explains 36.73% of the variation in incumbent vote share. The slope is positive. Our null hypothesis is that the population spending-gap slope is zero. Our alternative is that it differs from zero. The slope is positive ($t = 43.04$, two-sided $p < 0.001$), so we reject the null at 5%.

(b) Make a scatterplot of the two variables and add the regression line.

Answer: First, we calculate the fitted mean and its pointwise 95% confidence limits across the observed spending gaps (to use later in the scatterplot).

```
library(ggplot2)
dir.create("Figures", showWarnings = FALSE) # Create folder to store graphics.
spending_gap <- seq(min(incumbents$difflog), max(incumbents$difflog),
                    length.out = 200)
vote_grid <- data.frame(difflog = spending_gap)
vote_grid <- cbind(vote_grid, predict(vote_spending, vote_grid,
                              interval = "confidence", level = 0.95))
```

```
vote_spending_plot <- ggplot(vote_grid, aes(x = difflog)) +
  geom_point(data = incumbents, aes(y = voteshare),
             size = 0.8, alpha = 0.28, colour = "grey35") +
  geom_line(aes(y = lwr), colour = "red4", linetype = "dashed") +
  geom_line(aes(y = upr), colour = "red4", linetype = "dashed") +
  geom_line(aes(y = fit), colour = "steelblue2", linewidth = 0.7) +
  scale_y_continuous(labels = scales::label_percent(accuracy = 1)) +
  labs(title = "Incumbent vote share and campaign spending",
       subtitle = "Fitted mean in blue with dashed 95% confidence limits",
       x = "difflog (Log incumbent spending minus log challenger spending)",
       y = "voteshare (Incumbent vote share)") +
  theme_classic(base_size = 12)
ggsave("Figures/Q1_vote_spending.pdf", vote_spending_plot,
       width = 7, height = 4)
```



The spending gap explains 36.73% of the variation in incumbent vote share. Individual results vary much more widely around the fitted line than the confidence limits for the mean.

(c) Save the residuals of the model in a separate object.

Answer: We save the residuals and check the first election. A residual is the observed vote share minus the fitted vote share.

```
vote_resid <- resid(vote_spending)
first_election <- c(observed = incumbents$voteshare[1],
                    fitted = unname(fitted(vote_spending)[1]),
                    residual = unname(vote_resid[1]))
print(round(first_election, 6)) # Rounded to 6 decimals.
```

```
observed    fitted  residual
0.602361  0.602784 -0.000423
```

The first incumbent received 60.24% of the vote, about 0.042 percentage points below the prediction. This explains the negative residual.

(d) Write the prediction equation.

Answer: We take the intercept and slope from the fitted model:

```
round(coef(vote_spending), 6) # Rounded to 6 decimals.
```

```
(Intercept)     difflog
   0.579031    0.041666
```

For election i, the prediction equation is:

$$\widehat{\texttt{voteshare}}_i = 0.579031 + 0.041666\,\texttt{difflog}_i$$

At equal spending ($\texttt{difflog} = 0$), the predicted incumbent vote share is 57.90%. For a one-unit increase in the log spending gap, the model predicts an increase of 0.041666 in incumbent vote share, or 4.17 percentage points.

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

(1) Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

Answer: We will use `difflog` to predict the vote share of the presidential candidate from the incumbent's party (`presvote`).

```
pres_spending <- lm(presvote ~ difflog, data = incumbents)
summary(pres_spending)
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.507583   0.003161  160.60   <2e-16 ***
difflog     0.023837   0.001359   17.54   <2e-16 ***

Multiple R-squared:  0.08795,   Adjusted R-squared:  0.08767
```

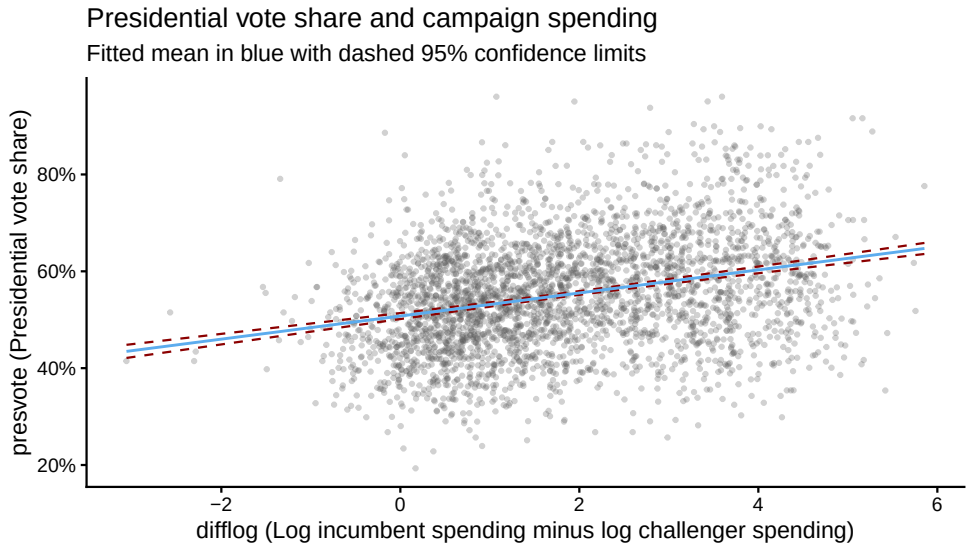
Our null hypothesis is that the population spending-gap slope is zero. Our two-sided alternative is that it differs from zero. We reject the null at 5% ($t = 17.54$, $p < 0.001$). The association is positive. The gap explains 8.80% of the variation in presidential vote share.

(2) Make a scatterplot of the two variables and add the regression line.

Answer: We put `difflog` on the horizontal axis and `presvote` on the vertical axis. We calculate the 95% confidence limits from the model in part (a).

```
pres_grid <- data.frame(difflog = spending_gap)
pres_grid <- cbind(pres_grid, predict(pres_spending, pres_grid,
                           interval = "confidence", level = 0.95))
```

```
pres_spending_plot <- ggplot(pres_grid, aes(x = difflog)) +
  geom_point(data = incumbents, aes(y = presvote),
             size = 0.8, alpha = 0.28, colour = "grey35") +
  geom_line(aes(y = lwr), colour = "red4", linetype = "dashed") +
  geom_line(aes(y = upr), colour = "red4", linetype = "dashed") +
  geom_line(aes(y = fit), colour = "steelblue2", linewidth = 0.7) +
  scale_y_continuous(labels = scales::label_percent(accuracy = 1)) +
  labs(title = "Presidential vote share and campaign spending",
       subtitle = "Fitted mean in blue with dashed 95% confidence limits",
       x = "difflog (Log incumbent spending minus log challenger spending)",
       y = "presvote (Presidential vote share)") +
  theme_classic(base_size = 12)
ggsave("Figures/Q2_pres_spending.pdf", pres_spending_plot,
       width = 7, height = 4)
```



The line slopes upward, but the points are more spread out than in Question 1. The spending gap tells us less about the presidential vote.

(3) Save the residuals of the model in a separate object.

Answer: We save the presidential residuals and check the first election:

```
pres_resid <- resid(pres_spending)
first_election <- c(observed = incumbents$presvote[1],
                    fitted = unname(fitted(pres_spending)[1]),
                    residual = unname(pres_resid[1]))
print(round(first_election, 6)) # Rounded to 6 decimals.
```

```
observed   fitted residual
0.526778 0.521173 0.005606
```

The first residual is positive: the presidential candidate received 0.56 percentage points more than predicted.

(4) Write the prediction equation.

Answer: We can read the two coefficients from the model:

```
round(coef(pres_spending), 6) # Rounded to 6 decimals.
```

```
(Intercept)     difflog
   0.507583    0.023837
```

The prediction equation is

$$\widehat{\texttt{presvote}}_i = 0.507583 + 0.023837\,\texttt{difflog}_i$$

At equal spending, the predicted share is 50.76%. A one-unit increase in the log spending gap raises the predicted presidential vote share by 0.023837, or 2.38 percentage points.

Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

(1) Run a regression where the outcome variable is `voteshare` and the explanatory variable is `presvote`.

Answer: We regress `voteshare` ~ `presvote`:

```
vote_pres <- lm(voteshare ~ presvote, data = incumbents)
summary(vote_pres)
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.441330   0.007599   58.08   <2e-16 ***
presvote    0.388018   0.013493   28.76   <2e-16 ***

Multiple R-squared:  0.2058,    Adjusted R-squared:  0.2056
```

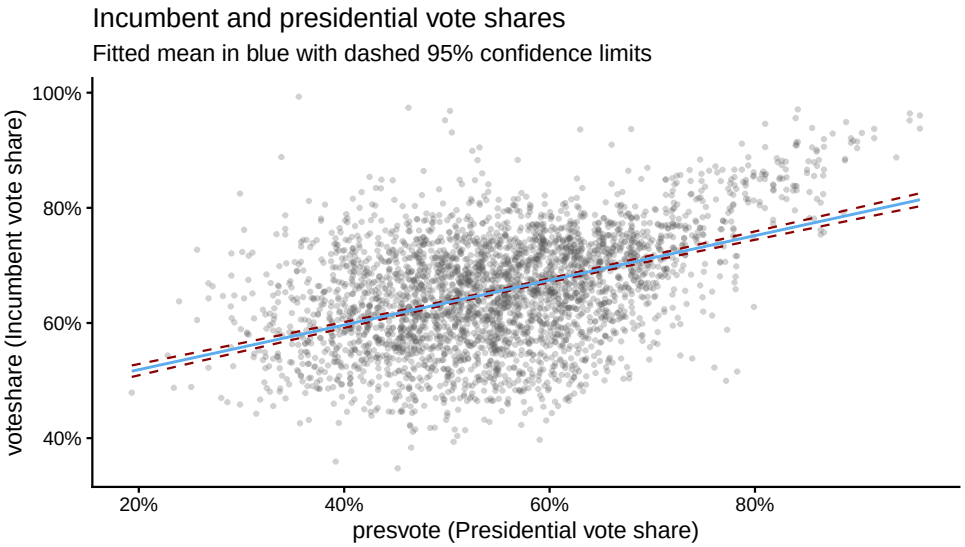
Our null hypothesis is that the population slope on presidential vote share is zero. Our two-sided alternative is that it differs from zero. We reject the null at the 5% level ($t = 28.76$, $p < 0.001$). Greater support for the party's presidential candidate predicts greater support for its incumbent. Presidential vote share explains 20.58% of the variation in incumbent vote share.

(2) Make a scatterplot of the two variables and add the regression line.

Answer: We plot one point per election and express both voteshares as percentages on the axes. We include a presidential vote share of 0.50 when calculating the CI.

```
pres_share <- seq(min(incumbents$presvote), max(incumbents$presvote),
                  length.out = 200)
support_grid <- data.frame(presvote = pres_share)
support_grid <- cbind(support_grid, predict(vote_pres, support_grid,
                                interval = "confidence", level = 0.95))
```

```
vote_pres_plot <- ggplot(support_grid, aes(x = presvote)) +
  geom_point(data = incumbents, aes(y = voteshare),
             size = 0.8, alpha = 0.28, colour = "grey35") +
  geom_line(aes(y = lwr), colour = "red4", linetype = "dashed") +
  geom_line(aes(y = upr), colour = "red4", linetype = "dashed") +
  geom_line(aes(y = fit), colour = "steelblue2", linewidth = 0.7) +
  scale_x_continuous(labels = scales::label_percent(accuracy = 1)) +
  scale_y_continuous(labels = scales::label_percent(accuracy = 1)) +
  labs(title = "Incumbent and presidential vote shares",
       subtitle = "Fitted mean in blue with dashed 95% confidence limits",
       x = "presvote (Presidential vote share)",
       y = "voteshare (Incumbent vote share)") +
  theme_classic(base_size = 12)
ggsave("Figures/Q3_vote_pres.pdf", vote_pres_plot,
       width = 7, height = 4)
```



Around a presidential vote share of 50%, we see incumbents who lost their race as well as incumbents who won comfortably.

(3) Write the prediction equation.

Answer: We use the two coefficients from the fitted line:

```
round(coef(vote_pres), 6) # Rounded to 6 decimals.
```

```
(Intercept)    presvote
   0.441330    0.388018
```

The prediction equation is

$$\widehat{\texttt{voteshare}}_i = 0.441330 + 0.388018\,\texttt{presvote}_i$$

If presidential vote share rises from 50% to 60%, the predicted incumbent share rises by $0.388018(0.60 - 0.50) \approx 0.03880$, or 3.88 percentage points.

Question 4

The residuals from part (a) tell us how much of the variation in `voteshare` is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in `presvote` is *not* explained by the difference in spending between incumbent and challenger in the district.

(1) Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

Answer: We can use the incumbent vote-share residuals saved in Question 1(c) as the outcome and the presidential vote-share residuals saved in Question 2(c) as the explanatory variable.

```
residual_data <- data.frame(vote_resid, pres_resid)
resid_model <- lm(vote_resid ~ pres_resid, data = residual_data)
summary(resid_model)
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.942e-18  1.299e-03    0.00        1
pres_resid   2.569e-01  1.176e-02   21.84   <2e-16 ***

Multiple R-squared:   0.13, Adjusted R-squared:  0.1298
```

Our null hypothesis is that the population slope relating the two spending-adjusted shares is zero. Our two-sided alternative is that it differs from zero. We reject the null at the 5% level ($t = 21.84$, $p < 0.001$). The presidential residual explains 13.00% of the variation left in incumbent vote share after adjusting for spending.

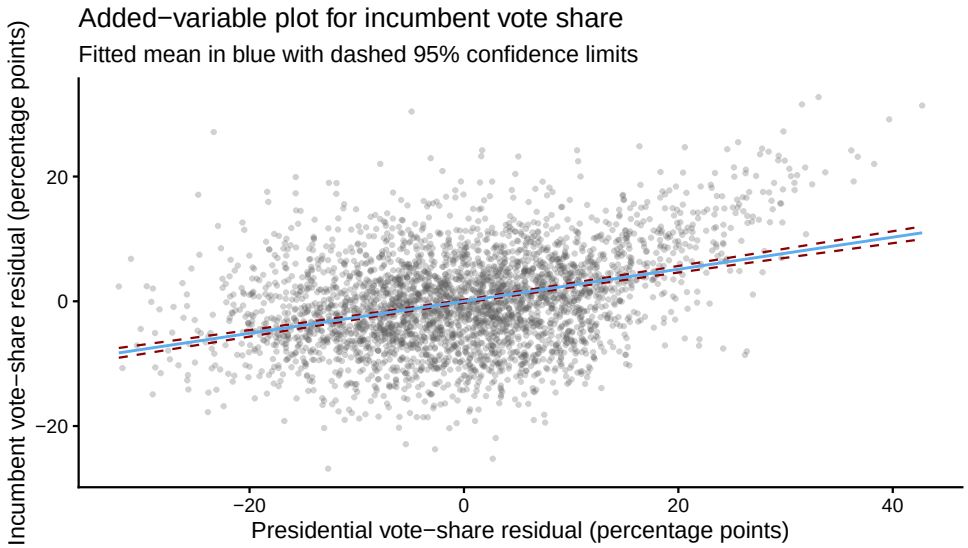
(2) Make a scatterplot of the two residuals and add the regression line.

Answer: We put the presidential residual on the horizontal axis and the incumbent residual on the vertical axis.

```
pres_gap <- seq(min(residual_data$pres_resid), max(residual_data$pres_resid),
                length.out = 200)
resid_grid <- data.frame(pres_resid = pres_gap)
resid_grid <- cbind(resid_grid, predict(resid_model, resid_grid,
                                interval = "confidence", level = 0.95))
```

```
resid_plot <- ggplot(resid_grid, aes(x = pres_resid)) +
  geom_point(data = residual_data, aes(y = vote_resid),
             size = 0.8, alpha = 0.28, colour = "grey35") +
  geom_line(aes(y = lwr), colour = "red4", linetype = "dashed") +
  geom_line(aes(y = upr), colour = "red4", linetype = "dashed") +
  geom_line(aes(y = fit), colour = "steelblue2", linewidth = 0.7) +
  # To read the residuals in percentage points, we multiply the axis labels by 100.
  scale_x_continuous(labels = scales::label_number(scale = 100)) +
  scale_y_continuous(labels = scales::label_number(scale = 100)) +
```

```
  labs(title = "Added−variable plot for incumbent vote share",
       subtitle = "Fitted mean in blue with dashed 95% confidence limits",
       x = "Presidential vote−share residual (percentage points)",
       y = "Incumbent vote−share residual (percentage points)") +
  theme_classic(base_size = 12)
ggsave("Figures/Q4_added_variable.pdf", resid_plot,
       width = 7, height = 4)
```



Each point represents one election. The presidential candidate outperformed the prediction from spending, while the incumbent underperformed it.

(3) Write the prediction equation.

Answer: We take the coefficients from the residual regression:

```
round(coef(resid_model), 6) # Rounded to 6 decimals.
```

```
(Intercept)  pres_resid
   0.000000    0.256877
```

$$\widehat{\texttt{vote_resid}}_i = 0.256877\,\texttt{pres_resid}_i$$

Each original regression includes an intercept, so both sets of residuals have mean zero. Their fitted line passes through the origin. When the presidential candidate exceeds the spending prediction by one percentage point, we predict an incumbent residual of +0.257 percentage points.

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

(a) Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

Answer: We will include the spending gap and presidential vote share in the same regression. Each slope measures the association with incumbent vote share while we hold the other predictor fixed.

```
full_model <- lm(voteshare ~ difflog + presvote, data = incumbents)
summary(full_model)
```

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.4486442  0.0063297   70.88   <2e-16 ***
difflog     0.0355431  0.0009455   37.59   <2e-16 ***
presvote    0.2568770  0.0117637   21.84   <2e-16 ***

Multiple R-squared:  0.4496,    Adjusted R-squared:  0.4493
```

For each predictor, our null hypothesis is that its population slope is zero with the other predictor held fixed. The two-sided alternative is that its slope differs from zero. We reject both null hypotheses at the 5% level ($p < 0.001$ for each slope). Together the predictors explain 44.96% of the variation in incumbent vote share.

(b) Write the prediction equation.

Answer: We use the fitted intercept and both partial slopes:

```
round(coef(full_model), 6) # Rounded to 6 decimals.
```

```
(Intercept)     difflog    presvote
   0.448644    0.035543    0.256877
```

Therefore, the prediction equation is

$$\widehat{\texttt{voteshare}}_i = 0.448644 + 0.035543\,\texttt{difflog}_i + 0.256877\,\texttt{presvote}_i$$

Holding presidential vote share fixed, a one-unit increase in the log spending gap predicts an increase of 0.035543 in incumbent vote share, or 3.55 percentage points. Holding the spending gap fixed, a rise from 50% to 60% in presidential vote share predicts 2.57 percentage points more for the incumbent. The corresponding unadjusted difference in Question 3 was 3.88 points.

(c) What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

Answer: The coefficient on `presvote` in Question 5 equals the coefficient on `pres_resid` in Question 4 to numerical precision. We compare the unrounded estimates and their standard errors.

```
slopes <- c(Q4 = unname(coef(resid_model)["pres_resid"]),
            Q5 = unname(coef(full_model)["presvote"]))
print(slopes, digits = 10)
print(all.equal(unname(resid(resid_model)), unname(resid(full_model))))
se <- c(Q4 = coef(summary(resid_model))["pres_resid", "Std. Error"],
        Q5 = coef(summary(full_model))["presvote", "Std. Error"])
print(se, digits = 8)
```

```
           Q4            Q5
0.2568770127 0.2568770127
[1] TRUE
           Q4           Q5
0.011761902 0.011763746
```

Both slopes equal 0.2568770127, and the residuals match to numerical precision. For a one-percentage-point increase in presidential vote share at the same spending gap, the model predicts about 0.257 percentage points more incumbent vote share.

For the explanation of why this happens, we can see the slides 12–14 of the Week 9 tutorial: In Question 1, we subtract the incumbent vote share predicted by spending from the observed incumbent share. In Question 2, we do the same for the presidential share, using the same spending gap and the same elections. What remains in each share is uncorrelated with the spending gap. Regressing the first residual on the second therefore uses exactly the variation that estimates the presidential slope when we hold spending fixed in Question 5. This is the Frisch–Waugh–Lovell result.

The standard errors both round to 0.01176. For the error variance, R divides the same residual sum of squares by 3,191 in Question 4 and by 3,190 in Question 5, where we estimate one more coefficient. This gives the slightly larger standard error in Question 5. The R^2 values use different totals as well. Question 4 accounts for 13.00% of the incumbent variation remaining after spending, and Question 5 accounts for 44.96% of the original variation.